Meric Altug Gemalmaz, Ph.D.

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EDUCATION

Purdue University, West Lafayette, Indiana

- Ph.D. in Computer Science (Human-AI Interaction Focused) **3.96/4.0** Jan. 2020 – June 2025
- M.S. in Computer Science (Transferred to Ph.D.) **3.93/4.0** Jan. 2019 – Dec. 2019
- B.S. in Computer Science (Software Engineering Track, **Distinction**) **3.92/4.0** Aug. 2015 – Dec. 2018

SKILLS

Programming Languages: Python, C/C++, C#, JavaScript, Java, HTML/CSS, R, SQL, Bash
Machine Learning: Large-scale retrieval and ranking, recommender systems, content-integrity and quality signals, LLM-based signal design and prompting, model evaluation, A/B experimentation and metric design
Machine Learning Tools: scikit-learn, GSL, NumPy, Pandas
Tools & Platforms: Git, Meteor.js, MongoDB, MTurk, Android SDK, Firebase, GDB

EXPERIENCE

ML Research Scientist, Meta, Menlo Park, CA Aug. 2025 – Present

- Work on Facebook Groups' large-scale ranking systems: content recommendation and content quality.
- In roughly nine months on Groups, shipped **17** A/B-tested launches across the group recommendation and unconnected group post recommendation stacks, building LLM-based content-quality signals, **two** new candidate generators, and content-integrity detection signals.
- Learned the group and unconnected post recommendation stacks in my first weeks, took production oncall by month **four**; chased root causes in retrieval and measurement instead of patching symptoms.
- In the Meta AI Org, create and improve training data for AI models, build the workflows that produce it, and evaluate AI agent systems.

Web App Development for Data Collection, Purdue Jan. 2020 – May 2025

- Coordinated **1,500+** human participants across **4+** research projects on **MTurk** for data collection and analysis.
- Developed and deployed web applications using **JavaScript** and **Meteor.js**, managed back-end databases with **MongoDB**, and implemented front-end interfaces using **HTML** and **CSS**.
- Delivered **90%** participant satisfaction and blocked **80%** of the bot attacks the system received by designing scenario-based, interactive UIs.

Fairness in AI-Driven Gig Work: User-Centric Insights, Purdue Dec. 2023 – May 2025

- Initiated a collaborative study with gig workers on the long-term impact of AI-driven gig assignments at varying fairness levels, yielding actionable insights for fairer algorithmic job management.

Design Implications of Fairness in Human-AI Interactions, Purdue [P1, P3] Mar. 2021 – Dec. 2023

- Conducted **six** human-subject experiments on how loan applicants' repeated interactions with an AI-based loan approval system affect their fairness perceptions and willingness to keep engaging with the AI.
- Simulated **1,000** loan applicant interactions using a **Markov Decision Process** in **Python** to estimate experiment parameters; collected real-world data and applied **regression analysis** in **R** to uncover key human-AI interaction trends.
- Found a critical fairness issue: over **75%** of users continue using biased models out of necessity rather than perceived fairness, urging developers to rethink usage metrics as indicators of model fairness.

Data Bias Mitigation Algorithm, Purdue [P2] Mar. 2020 – Mar. 2021

- Leveraged unsupervised learning techniques to detect and mitigate cognitive bias in crowdsourced data annotations.
- Applied **probabilistic graphical models** and **Expectation-Maximization** with the **GSL** library in **C** to reduce annotator bias, improving label accuracy across **100** simulated and **12** real-world tasks.
- Achieved over **10%** increase in inferred label accuracy over existing baselines through reduction in annotation bias.

HONORS AND AWARDS

Recipient of the Graduate Teaching Award, Purdue Nov. 2022
Recipient of the Raymond Boyce Graduate Teaching Award, Purdue Apr. 2020
Dean's List and Semester Honors (**8 Semesters**), Purdue Aug. 2015 – Dec. 2018

SELECTED PAPERS

[P1] Meric Altug Gemalmaz, Ming Yin. "Understanding Decision Subjects' Fairness Perceptions and Retention in Repeated Interactions with AI-Based Decision Systems." *Proceedings of the 5th AAAI/ACM Conference on AI, Ethics, and Society (AIES)*, Oxford, UK, Aug. 2022.

[P2] Meric Altug Gemalmaz, Ming Yin. "Accounting for Confirmation Bias in Crowdsourced Label Aggregation." *Proceedings of the 30th International Joint Conference on Artificial Intelligence (IJCAI)*, Online, Aug. 2021.

[P3] Meric Altug Gemalmaz, Ming Yin. "Understanding Decision Subjects' Engagement with and Perceived Fairness of AI Models When Opportunities of Qualification Improvement Exist." *arXiv:2410.03126*